

Neural Operators for Disruption Prediction in Nuclear Fusion Plasmas



Sreeyutha Ratala

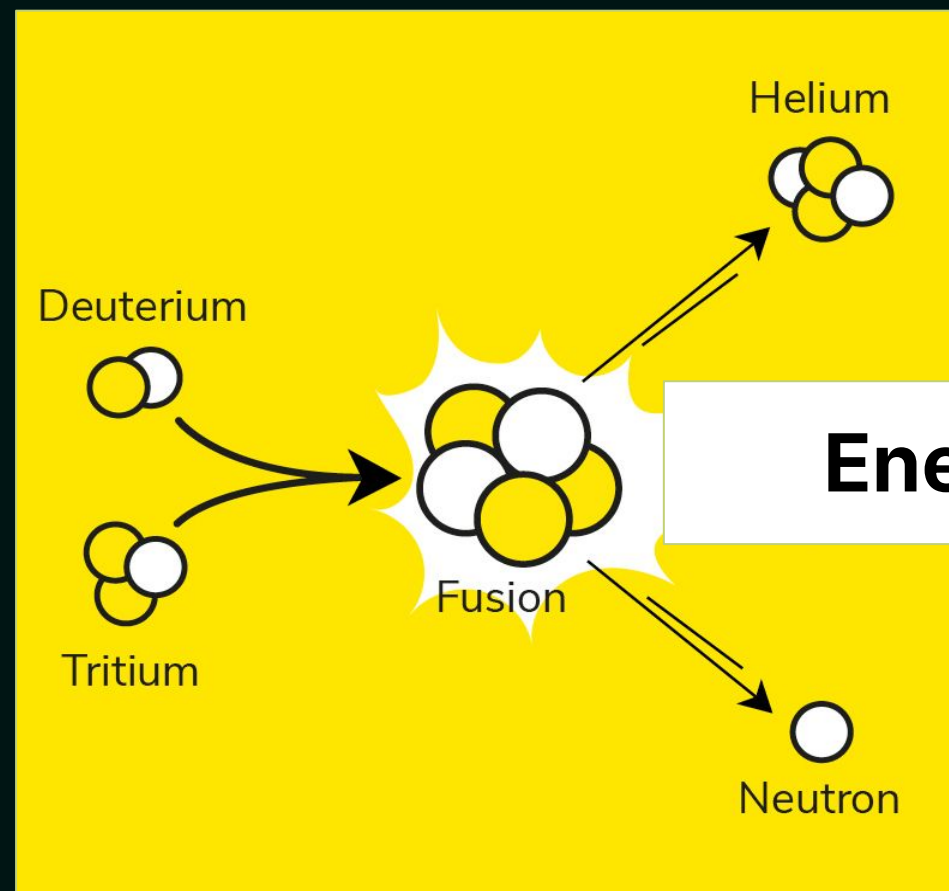
Mentor: Valentin N. Duruisseaux

Prof. Anandkumar

Introduction.

Nuclear Fusion is the solution to our growing energy and climate needs.

2 atomic nuclei combine



Energy



Plasma
powerful & promising
clean, safe, limitless



need:
>>100 million °C
pressure
confine plasma
(high instabilities)

the problem?
don't have
sun's gravity
to spark
fusion

But it's hard to control.

Symmetric...BLIT

Disruption Prediction

Machine Learning

PLASMA
ELECTRICAL
CURRENT

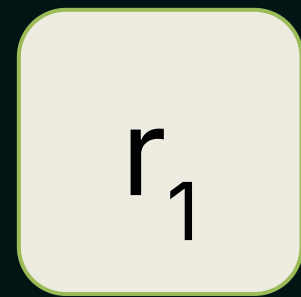
TOROIDAL
MAGNETIC
FIELD

HELICAL
MAGNETIC
FIELD

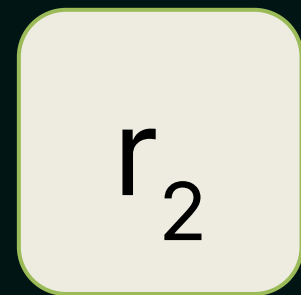
Method.

Data

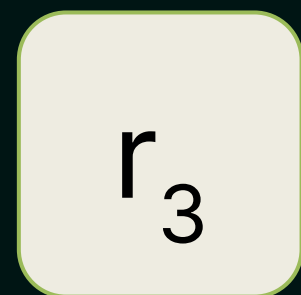
Data Generating Tokamaks



MIT
C-Mod



General Atomic
DIII-D



Chinese Academy
EAST

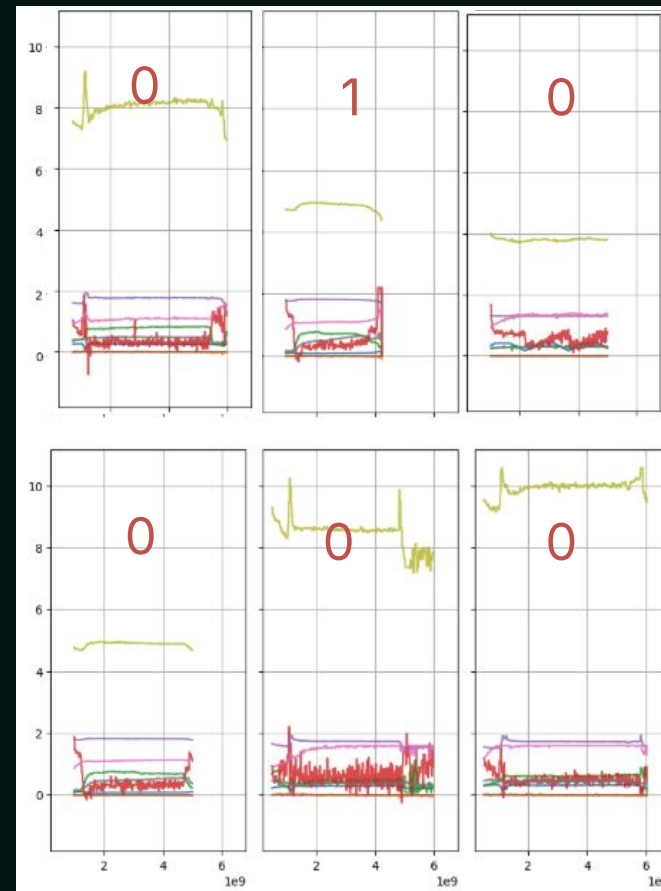
sampling rates
(ms)



Plasma State Observations

0: Non-Disruptive
1: Disruptive

imbalanced
5:1



12 continuous features
22,556 shots/trials
Sequence runs for seconds to minutes
Clean Nans

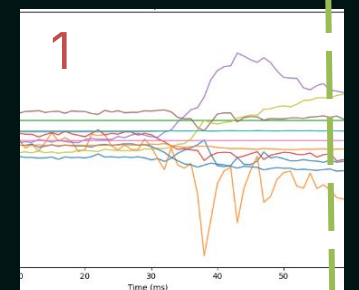
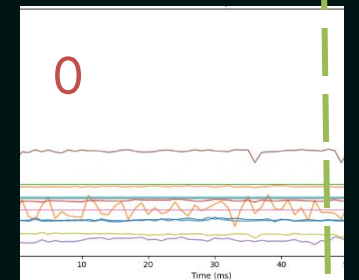
Embedding Vectors

concatenate all
features

$$h_a^{(1)}, \dots, h_a^{(t)}$$

$$h_b^{(1)}, \dots, h_b^{(t)}$$

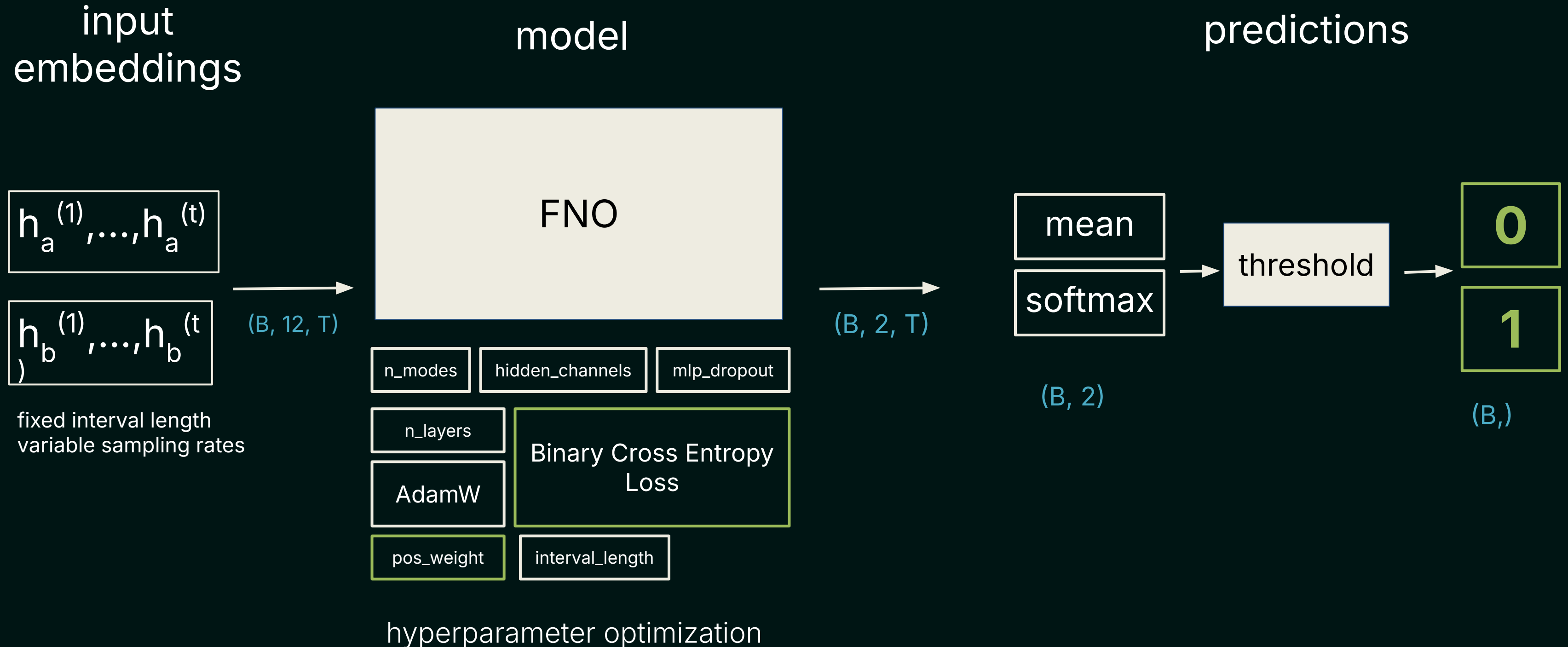
Normalize



⋮
cutoff portion
of disruptive
interval

clip data at 1st & 99th
percentiles

Fourier Neural Operator (FNO)

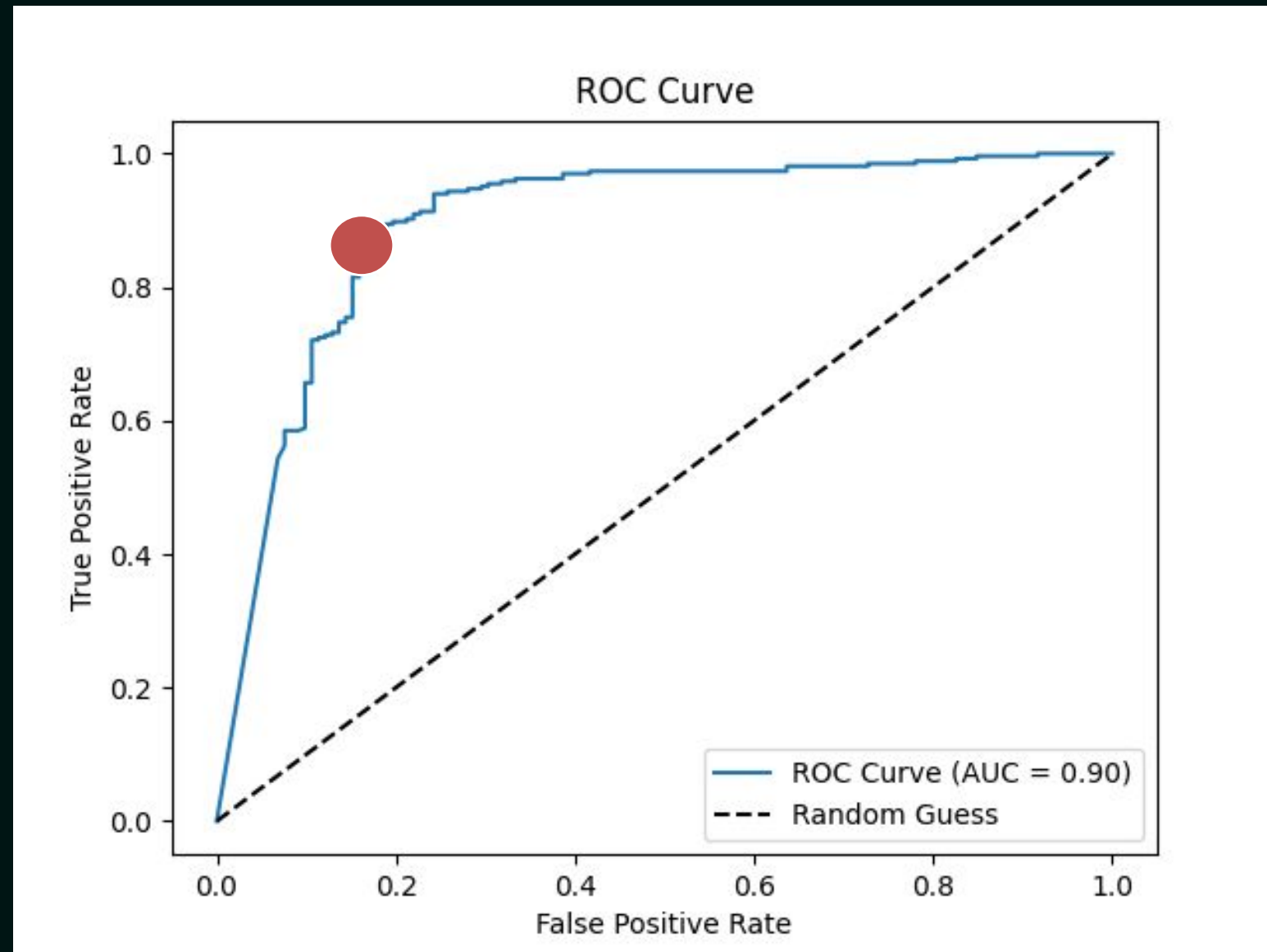


Results.

Overall Metrics (Test set)

		predicted			
		disruption	non disruption		
actual	disruption	predict disruptions when they occur	missed a disruption: can severely damage tokamak	90%	predict disruptions when they occur
	non disruption	false alarms on healthy plasma discharges	lack of disruption successfully detected	10%	don't predict disruptions when they do occur

Overall Metrics (Test set)



90% chance model ranks
random disruption > random non-disruption

More important to classify
every disruption than to
minimize false alarms.

0.86 accuracy

0.934 AUPRC

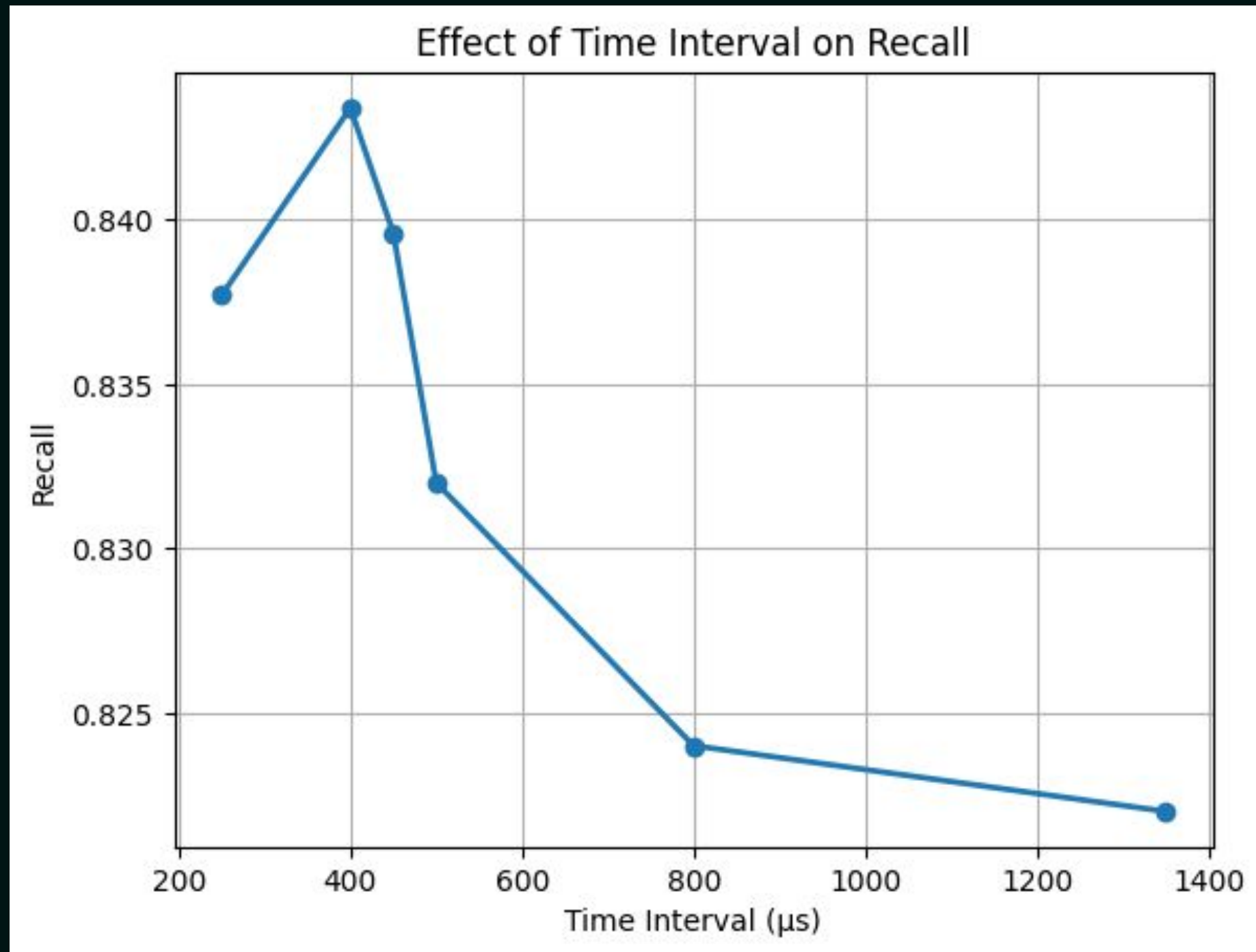
0.864 F1

0.90 AUC

Sensitive to
class
imbalance

[0, 1]

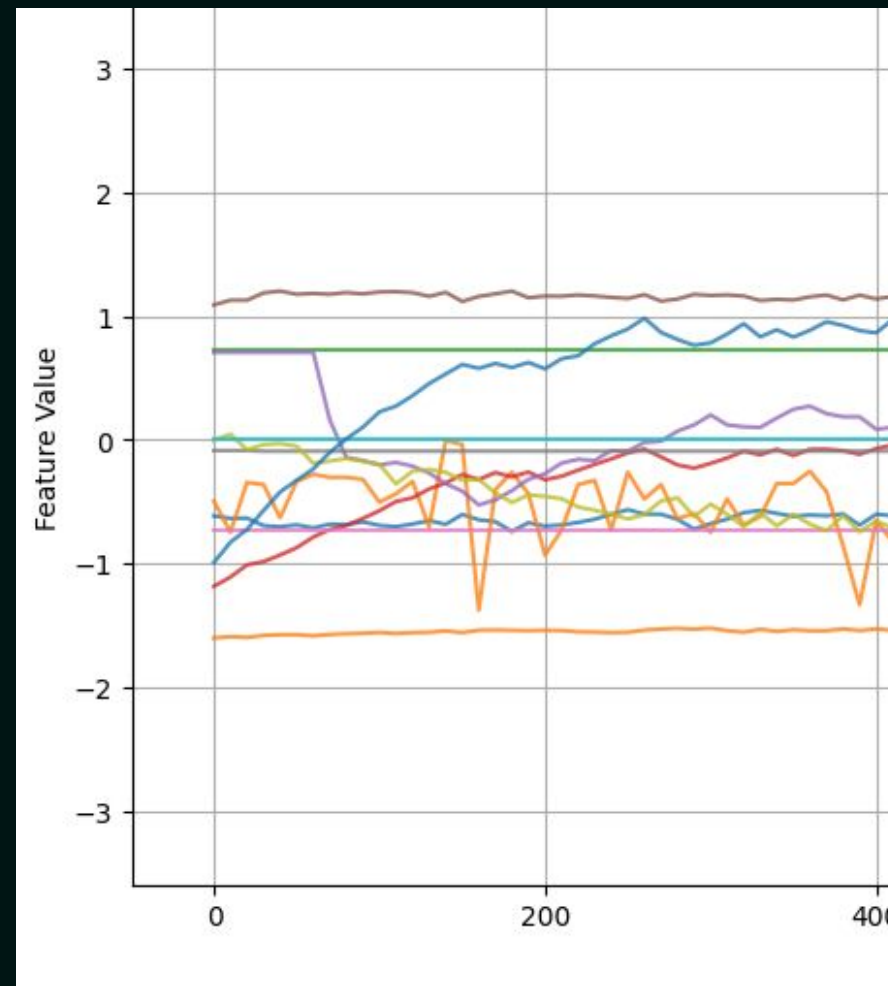
Longer Time Intervals Improve Recall (Up to a Point)



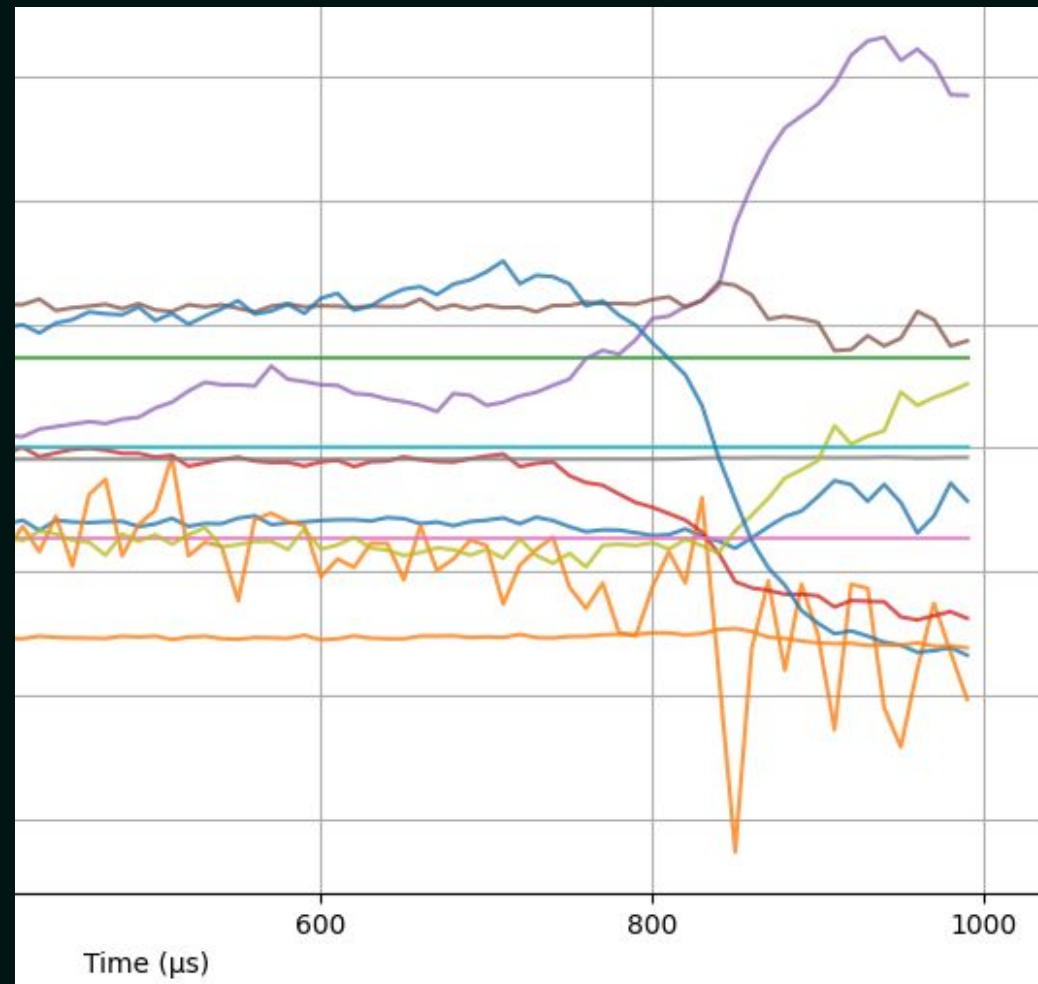
Proportion of disruptions correctly predicted **increase** with longer time intervals up to **~400 microseconds**, at which recall goes down and model doesn't predict as much of the disruptions.

Future Work.

Predict next sequence, then classify



a



b



disruption

Incorporate disruption prediction algorithm into **disruption mitigation system** such as ITER nuclear fusion experimental facility

Questions?